Suraj Mishra

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Professional Summary

Software engineer focused on GPU kernel optimization and LLM inference, with nearly four years of professional software development at Oracle and Incture and a master's from NYU. Built Triton attention and KV-cache kernels validated by 210 H100 tests and authored merged vLLM/SGLang contributions to serving observability and benchmark correctness. Combines GPU profiling, reproducible performance measurement, backend development and infrastructure automation to make inference faster and more dependable.

Education

New York University **New York, NY**
Masters in Computer and Information Systems | GPA: 3.72/4.00 *Sept 2024 - May 2026*

- **Coursework:** Deep Learning, Machine Learning, Building LLM Reasoners, Operating Systems

National Institute of Technology, Bhopal **Bhopal, India**
B.Tech in Computer Science and Engineering *Jul 2016 - Jun 2020*

Work Experience

Oracle **Jan 2022 - Aug 2024**
Member Technical Staff *Bengaluru, India*

- Redesigned **JUnit 5/Jupiter test execution and object lifecycles**, reducing end-to-end execution of the then-existing full pipeline from **approximately 10-11 hours to within 3 hours**; enabled expansion into functional, load, stress and long-running suites.
- Moved dependent Recovery Appliance, policy, network and protected-database interface objects into shared setup; separated verification from state-changing tests, grouped compatible tests for parallel/sequential reuse and distributed suites across compute instances through **TeamCity**.
- Adapted existing **Terraform** provisioning scripts for tenancy/network settings; authored readiness and artifact-collection scripts plus cleanup that ran despite earlier failures, and deployed development code/images with **Ansible**.
- Automated Recovery Appliance migration test workflows from provisioning through teardown, checking recovery windows, data-loss exposure, protected states and backup continuity; expanded specification-based functional coverage on representative hardware.
- Researched Java logging libraries for automatic test-class/function attribution and wrote a shell script organizing interleaved output into class-based logs to investigate service error-code, message and resource-field discrepancies.
- Tested **Oracle Database Zero Data Loss Autonomous Recovery Service** in Azure and Google Cloud environments.
- Validated new rack and bare-metal hardware with point-in-time recovery and functional tests; documented procedures and partially automated them for TeamCity integration.

Incture Technologies **Sept 2020 - Dec 2021**
Software Engineer *Bengaluru, India*

- Built stateless **Java/Spring Boot** backend microservices for enterprise supply-chain analytics supporting **5× traffic growth**; reduced API **p99 latency by about 70%** through SQL plan profiling, indexing and connection-pool tuning.
- Led root-cause investigations into data consistency failures and orchestration deadlocks; automated **Jenkins/Ansible blue-green deployments**, eliminating release downtime and reducing customer-reported incidents by **about 85%**.

Projects

TritonAttn: Attention and KV-Cache Kernels | ***Triton, CUDA, PyTorch, H100*** **2026**

- Built a from-scratch GPU attention library reproducing FlashAttention-2, Flash-Decoding, GQA, StreamingLLM and KIVI as fused Triton kernels across prefill and decode; **210/210 tests passing on H100 (sm_90)**.
- Engineered a FlashAttention-2 prefill kernel reaching **433 TFLOP/s BF16** and **1.1-1.3× PyTorch SDPA at long sequence lengths** through online-softmax tiling, base-2 exp2 and Hopper autotuning.
- Implemented split-KV Flash-Decoding with an occupancy-aware split policy and tensor-core GQA/MQA path; achieved **up to 13.6× versus naive decode** and **peak 1.29× versus SDPA in target GQA regimes**.
- Implemented KIVI INT8/INT4/INT2 KV-cache quantization with in-kernel bit-unpacking; achieved **5.2× KV compression** and **+0.33% INT4 perplexity**, with bit-for-bit validation against dequantize-then-attend.
- Used **NVIDIA Nsight Compute** to diagnose register-pressure-capped occupancy; integrated kernels into Hugging Face Transformers and validated token-identical greedy generation.

Merged vLLM Contributions | ***Python, async HTTP, pytest, model serving*** **Sept 2026 - Present**

- Authored merged **vLLM #55508**, correcting time-to-first-token (TTFT), end-to-end latency and time-per-output-token (TPOT) accounting across completion, chat, audio and legacy inference benchmark clients.
- Unified the first-token timestamp, stopped non-token usage trailers from extending request latency and rejected usage-only streams; enforced the decode-window timing invariant with **12 new and 8 sibling tests passing**.
- Authored merged **vLLM #62205**, adding request-ID-correlated completion output logging for streaming deltas, non-streaming responses and final accumulated output; validated **53 focused and 281 broader regression tests**.
- Authored merged **vLLM #62204** to document and regression-test remote-media retry byte-budget accounting across synchronous/asynchronous downloads; improved operator-facing guidance with **9 focused tests passing**.

Merged SGLang Scheduler Metrics Fix | ***Python, request lifecycle, Prometheus*** **Sept 2026**

- Authored merged **SGLang #39312**, reproducing negative queue-time metrics and a scheduler assertion when retracted or preempted requests re-entered the waiting queue.
- Accumulated every wait interval, closed waits idempotently across chunked prefill and preserved values across process-clock rebasing; retained scheduling decisions and the existing Prometheus histogram contract.
- Added six regression/contract tests; validated **303 related unit tests and 20,000 randomized lifecycles** across unified/disaggregated modes, preserving behavior on **9,504 non-retracted cases**.

LLM Inference Control Plane | ***Go, Kubernetes, Kubebuilder, Prometheus*** **Aug 2026 - Present**

- Built a Kubernetes operator deploying self-hosted LLMs as an OpenAI-compatible service; gated canary promotion, rollback or hold on measured serving quality while keeping the last good revision available.
- Implemented an SSE sidecar measuring TTFT, errors, concurrency and queue depth; separated queue/concurrency-based autoscaling from rollout decisions and packaged dashboards with multi-window SLO alerts.
- Established p95 TTFT stability in a CPU feasibility study before using it as a release gate; hardened the operator with admission checks, deterministic tests and CI checks for generated API/documentation drift.

Technical Skills

Languages and GPU: Python, C++, Java, Go, SQL; Triton, CUDA, H100/A100, tensor cores, FP16/BF16, shared/global/L2 memory, occupancy and register-pressure analysis
Inference optimization: FlashAttention, split-KV decoding, GQA/MQA, INT2/INT4/INT8 KV-cache quantization, fusion, autotuning, Nsight Compute/Systems, CUDA events, roofline analysis
LLM systems: PyTorch, Transformers, vLLM, SGLang, llama.cpp, TensorRT-LLM, transformer architecture, TTFT/E2E/TPOT, continuous/dynamic batching, async HTTP, SSE
Training and testing: FSDP, DeepSpeed, NCCL, RDMA, GRPO, LoRA/QLoRA, W&B; pytest, JUnit 5, timing-invariant and randomized lifecycle testing, reproducible benchmarking
Infrastructure: Kubernetes, Kubebuilder, Docker, Terraform, Ansible, SLURM, Prometheus, Grafana, SLOs, Git/GitHub Actions, TeamCity, Jenkins, canary and blue-green delivery